

Introduction

For Pokémon Players

Cyrus will be a Pokémon TCG-playing software agent that utilizes machine learning to become a user's personal testing group, ready to simulate the hundreds of games that it takes to really test a Pokémon TCG deck against the entire range of matchups that one may expect to face at a major tournament. Equipped with the power to play the game at a level equal to or above the skill level a top 100 Masters player, Cyrus could efficiently simulate hundreds of different matchups by playing against itself and recording the results. Similar to how the use of an analytical spreadsheet helps players make mathematically-supported decisions on what to play in a tournament, Cyrus would be able to quickly simulate many games and determine the best deck to play for a tournament. While an analytical spreadsheet's numbers are all based off of games manually played and players' intuition, the statistics that Cyrus generates will help remove biases from our data and also see how small adjustments to certain decks affect its matchups in a massive field.

By just plugging in a variety of different decks, Cyrus will be able to generate data and statistics from the many games that it simulated in each matchup. The program will make it much easier for players to test their rogue ideas out quickly, see how a tech card could change a deck's performance against the metagame, and discover new strategies that could be used to turn an unfavorable matchup into a favorable one.

For AI Researchers

The *Pokémon Trading Card Game* (often referred to as *Pokémon TCG*) is a two-player zero-sum stochastic imperfect information game that is played with collectible cards. Its closest siblings are games such as *Magic: The Gathering*, *Hearthstone*, and *Yu-Gi-Oh!*. Unlike other imperfect information card games that AI systems have recently tackled, such as poker, *Pokémon TCG* players design their own custom decks from a pool of over 6,000 cards. Each player's deck must contain exactly 60 cards with no more than 4 copies of a single card. These cards vary in type, value, in-game applications, and strength, forcing players to carefully craft their decks to both execute its strategy consistently and exploit the weaknesses of common strategies (often referred to as "archetypes" or "meta decks").

The deck customization factor not only increases the sample space of possible game states and starting configurations to an impossibly large number, but also forces players to utilize very different strategies based on the composition of both players' decks. A good analogy would be a game of Chess where both players still play with a 20 piece army, but there exist a larger set



of pieces, each of which have different capabilities, that each player may choose from before the match to form their own 20-piece army. The starting positions of the pieces are then randomized, which results in the number of possible starting configurations increasing from 1 to $(\text{num pieces } C_{20})!$ for a single player. With over 6,000 cards legal in the Expanded Format, and up to 4 copies of each card allowed in one player's chosen 60 card deck, the feasibility of mastering the entire game in the same manner one would approach Chess or Go quickly fades.

Furthermore, Pokémon TCG requires players to make many moves in a single turn, the number of which is indeterminate from an observer's perspective. A player may want to make no moves (a simple "draw, pass") or could play enough cards to draw their entire deck. Players also need to consider the order in which they make specific moves in order to maximize the odds to find cards they need, draw certain cards off random effects, or avoid discarding valuable resources by chance (this is often referred to as sequencing).

Overall, Pokémon is a game filled with general skills (sequencing, resource conservation, etc), deck-specific skills (damage manipulation, Energy allocation, extent of knowledge of opponents resources, etc), and sometimes a mix of both. For a human player with a high level of general skill, acquiring deck-specific skills isn't all too difficult, but there is always a learning curve that may require 1-2 games to overcome. For a machine, learning curves like this often take a lot more training than a human would. Thus, our goal is to develop an AI system to:

- Play many different types of decks without extensive training
- Properly handle sequencing
- Quickly learn how new cards interact with old cards upon release
- Utilize multiple different sub-skills in order to generalize the overall skill (playing the game)
- Do all of this efficiently and at the level of a top 50 Masters Division player



Generalization

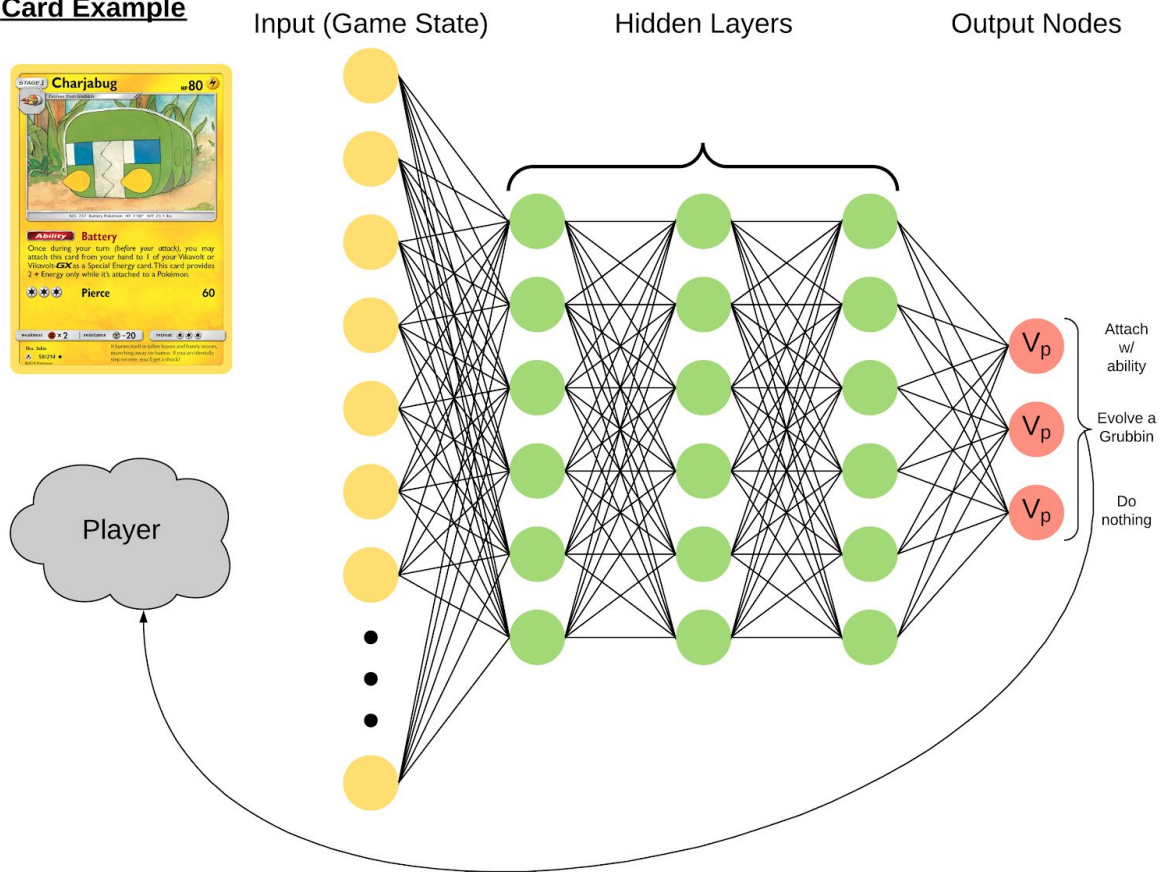
A major limitation of many AI systems that exist right now is the inability to handle diverse problems. While a Go or Chess-playing AI can handle many different board states well, every game of Go or Chess starts with the same pieces. Even AI systems built to play imperfect information games, such as DeepMind's AlphaStar, worked around the generalization problem by creating many different "agents" (players that have their own unique biases in order to play different characters). Pokémon is both an imperfect information game and an "adaptive strategy game" (a term we came up with to describe a game where both players' starting configurations (decks in Pokémon) are different from game to game), and to succeed in tournaments, a player must be prepared to adapt to new strategies on an event-to-event basis. In order to continue generating useful data as the game evolves and mitigate the need for retraining of the AI every time a new deck is tested, a player must be generalized; prepared to play many different decks at a high level and anticipate opponents' strategies based on common deck building trends in the game.

Many advanced AI systems, including AlphaGo and AlphaZero, are built only using two neural networks; one for decision-making, and one for game winner prediction. These networks are great at doing one thing each-either making a decision for the single move in a perfect information game or predicting the outcome by simulating opponents' moves. In order to generalize the player and equip it to play many different decks at the highest level, a hierarchy of many different neural networks must be established. Similar to how an administrator must make informed decisions based on information and input from those below them, the lower-level neural networks will provide information and opinions, based on their own biases, to the higher-level network that inevitably makes decisions. This can be referred to as the Board of Directors approach.



At the lowest level, every single card in the game will become its own neural network. Just like the pieces in a game like Wizard Chess or an advisor/intern to a higher-ranked official, the pieces will aid the player in making the best decision for the current situation in a game. Pokémon cards may be requesting that energy are attached to them more often because of their attack power, trainer cards may be pushing the player to use them in order to find a Pokémon or draw more cards that can aid the player further, and energy cards will cautiously advise the player as to where they believe their value could be used best.

ANN for Individual Card Example



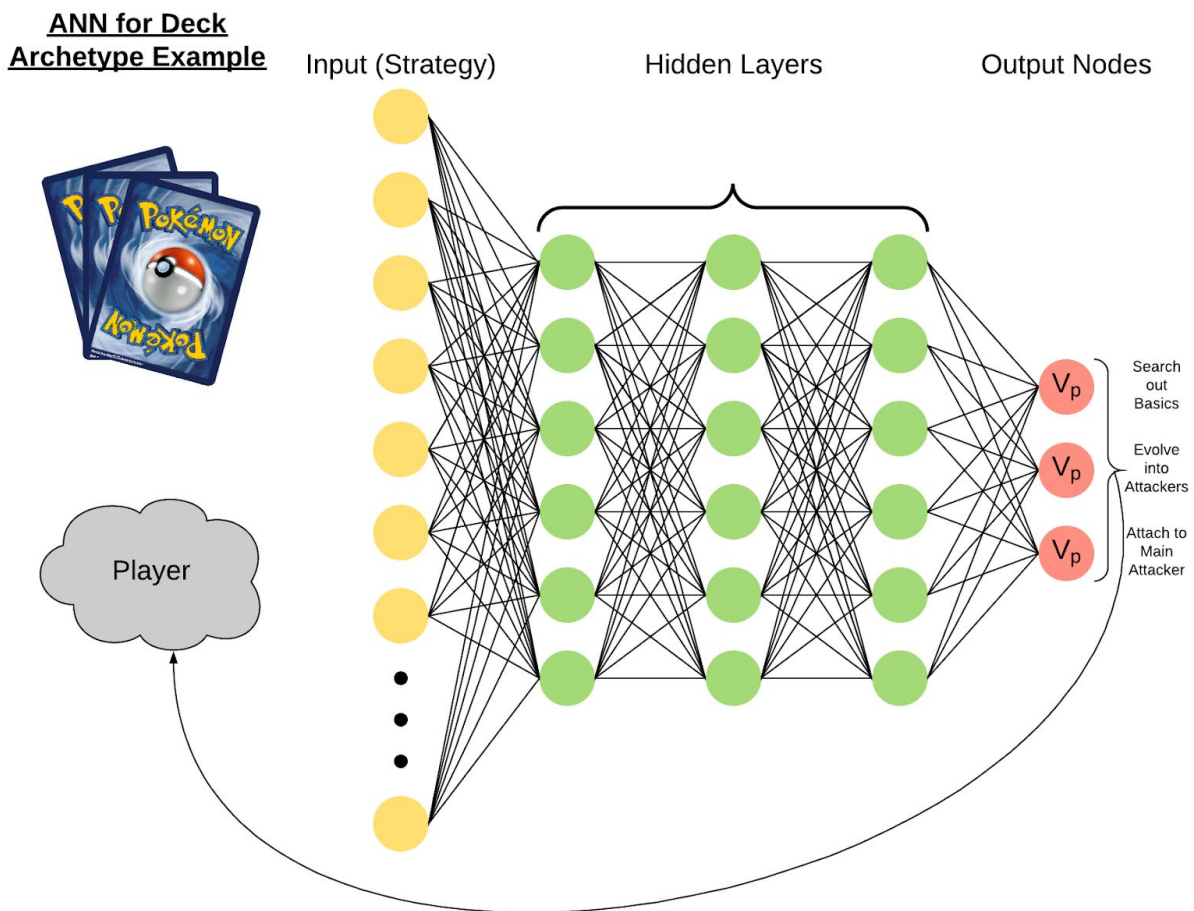
Above: a diagram of how a certain card's individual neural network could be constructed

The individual neural network that each card has returns an array of values, known as V_p , which represents the potential value that each possible use of the card represents. These values are then used as weights in the player network, allowing the player to tend toward different moves when playing different decks.



At the next level, the deck (collection of 60 cards) will develop its own biases as to how it should try to set up, how it should play differently against a specific opposing deck, and how it should try to win most of the time. These biases will influence how the player decides to play the deck at a more general level, rather than directly influencing each decision on every turn. For example, decks that aim to achieve different win conditions (e.g. take all six Prize Cards vs. run the opponent out of cards) would have very different biases, but decks that have the same general strategy, regardless of what Pokémon are in the deck as vehicles for executing said strategy, would have similar biases.

Some of the deck's biases can be determined by a few human inputs, such as most common win condition, main attacking Pokémon, and more. Think of this as the “voice of reason” that keeps the “selfish” cards in check and helps the player make better decisions.



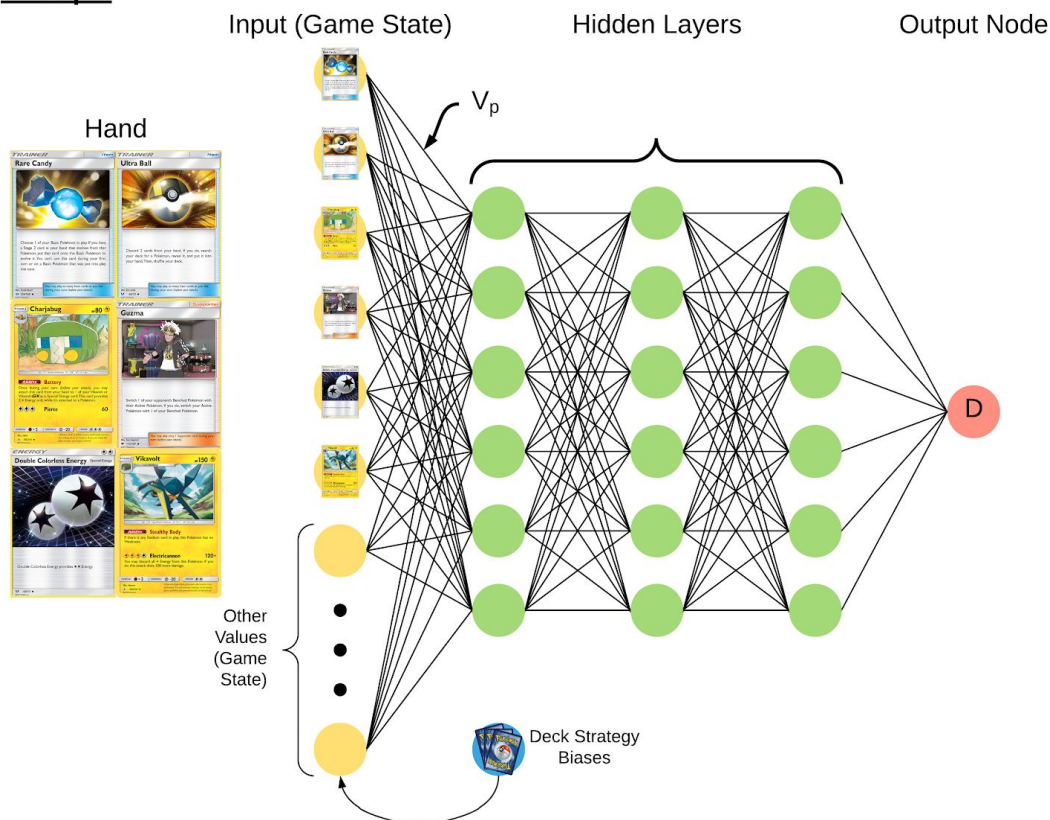
Above: a diagram of how a certain card's individual neural network could be constructed



Lastly, the player itself will weigh the inputs that it gets from the lower level neural networks in order to finally make a decision. It will take into account the current game situation as described by the players (e.g. attach this Energy or search for this Pokémon in order to attack and Knock Out that Pokémon), weigh the interests of the deck's overarching strategy (e.g. don't worry about attacking yet, focus on setting up the full strategy first), and consider what the opposing player's situation appears to be before making a final decision. Overall, this approach should be able to greatly simplify the generalization problem and make the implementation of a diverse and adaptable player possible.

The player will be trained to play given any deck; the difference in strategies for each deck are handled by the lower level networks that "report" information to the player in order for it to make informed decisions. The player has general skills and the cards below it provide the player with information necessary to play optimally given any game state.

ANN for Player Example



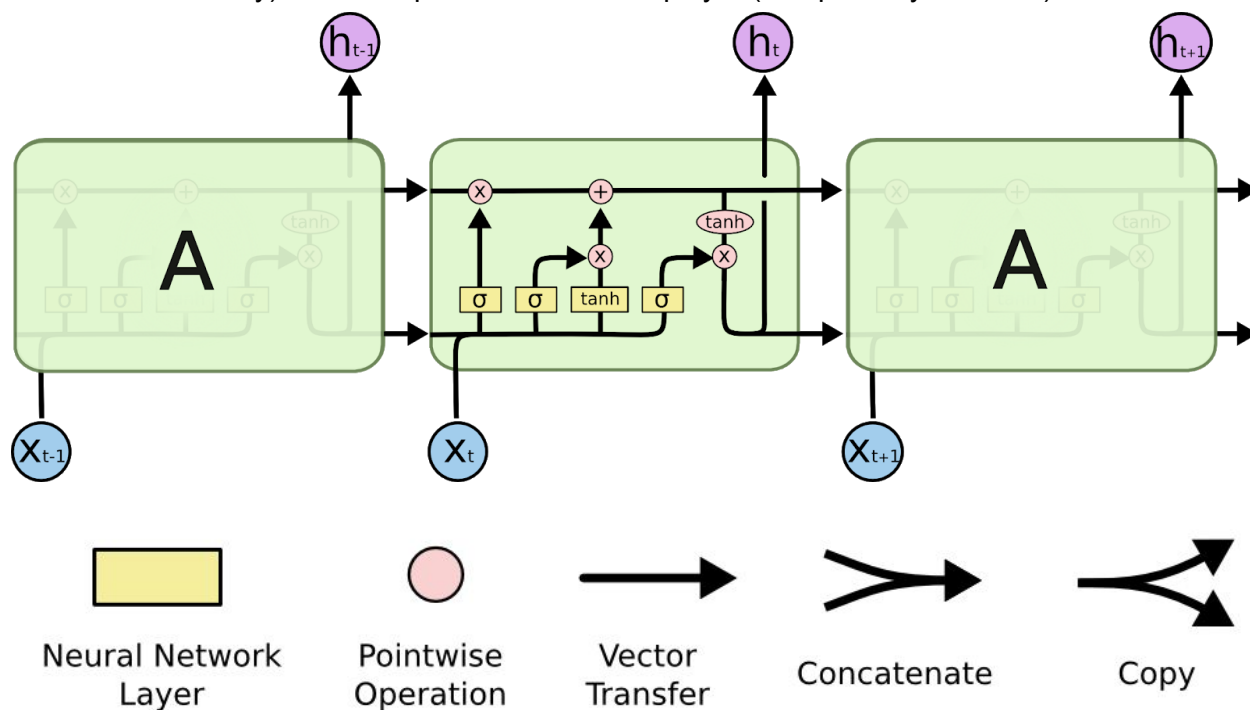
Above: a diagram of how the player's neural network would take input from each card's decisions in order to make its own decision. Diagram is inaccurate; the output should be n nodes, where n is the number of possible actions a player can take.



Multi-Move Turns

Pokémon TCG is a game in which players make many moves on their respective turns, the order of which often matters for optimizing the odds that certain moves pay off. Players must think of their turn as a string of moves that terminates when specific conditions have been achieved. Since the payoff of some moves directly depends on whether or not certain “imperfect information events” occur, such as when cards are randomly drawn, opponents make certain decisions, and coins are flipped, the entire string of moves is rarely determinable at the beginning of one’s turn. Therefore, since no Turing machine can calculate the entire string of moves a player must make on any given turn, moves must be calculated either individually or in separate sub-strings that divide where imperfect information events occur.

In order to aid in calculating these sub-strings of actions that a player may make, a player needs to take note of all that has happened in a game, cards they have played during the current turn, actions that may combine with other actions yet to be made, and the odds of drawing further cards that can assist a player. This requires strong memory, both long-term (game-wise) and short term (turn-wise). In order for an AI system to overcome this challenge, LSTM (Long Short-Term Memory) can be implemented into the player (and possibly the deck) level network.



Above: a basic diagram of an LSTM network (source: colah’s blog)



Handling Sequencing

A problem that often occurs in games of *Pokémon* is deciding the order in which to play two cards in situations where the order you play them in doesn't appear to matter. For example, a player may be searching for a specific Trainer card in their deck, but their only chance of getting this Trainer card is by using Jirachi's Stellar Wish ability; which allows a player to look at the top 5 cards of their deck and put a Trainer card into their hand. Say that player has 20 cards remaining in their deck and 1 copy of said Trainer card left in that 20-card deck. However, that player also plans to play an Ultra Ball later during their turn in order to search for a Pokémon and put it into their hand. Since the actions of playing Ultra Ball and utilizing Stellar Wish have no obvious impact on one another, the order that a player plays them in does not appear to be relevant.

When digging into the math, the odds of finding that Trainer card off of a Stellar Wish are 5/20 if the player uses Stellar Wish and then searches for a Pokémon with Ultra Ball. However, if the player uses Ultra Ball first, searches for a Pokémon, and then uses Stellar Wish, the odds of finding said Trainer card are now 5/19 as a Pokémon was removed from the deck with Ultra Ball's effect.

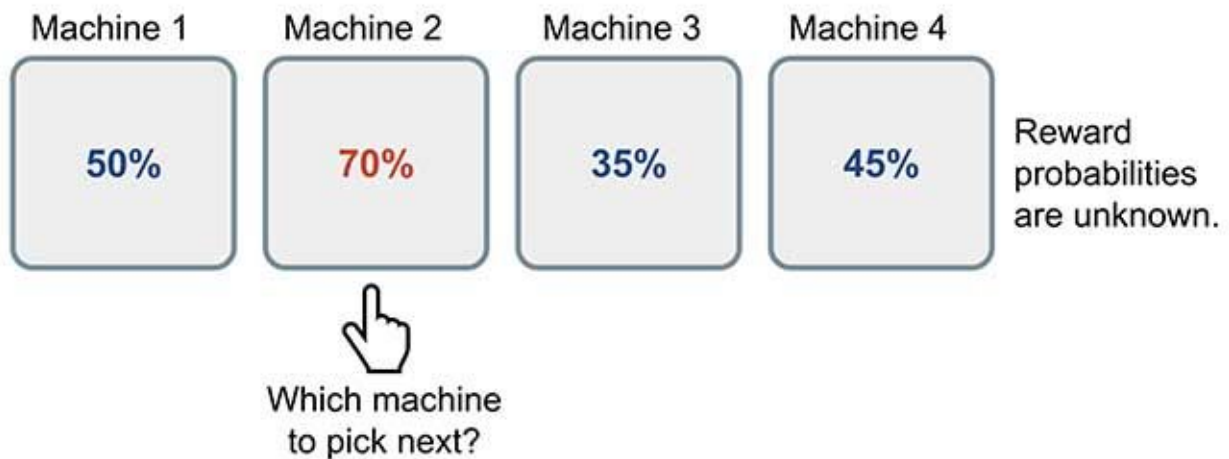
In tournaments, professional players are careful to correctly sequence their plays in order to maximize the odds of success. However, since these differences in odds are negligible and not immediately apparent, an AI may struggle to learn correct sequencing, resulting in inconsistent and imperfect play.



Above: Jirachi and Ultra Ball cards mentioned



This situation can be analogized to the multi-armed bandit problem; the card sequencing can be seen as the order in which the gambler plays each slot machine in order to maximize the overall odds of winning. There are many approximate solutions to the multi-armed bandit problem that likely will resolve sequencing well enough to handle simple situations.



Above: illustration of multi-armed bandit problem in the context of slot machines



Resource Conservation/Management

Another incredibly important skill top *Pokémon* players possess is the ability to think many turns ahead and conserve certain resources for later in the game. Due to the fact that Trainer cards are discarded when played and there are few effects that recover cards from the Discard Pile, almost all decks play each game with a limited number of resources. Often, certain cards are much stronger at specific points in a game, whether it be the early game, mid game, late game, or all the time. Incredibly strong cards are often included as 1-ofs (only 1 copy of the card is in the deck) and need to be conserved for later in the game, while other powerful cards are included as 3 or 4-ofs. While it may be tempting to play these powerful cards when they appear in a player's hand, it is often incorrect to overextend and use resources that are not necessary for the player to progress towards a win condition. Placing value on important cards and using them at the right time, sometimes even sacrificing the ability to play other cards in order to conserve these valuable resources, is a skill that separates novice players from the best in the world.

In order to tackle this, a combination of both LSTM and the Board of Directors approach can be implemented in order to place value on resources, consider the situations they are best used in, and inevitably make decisions on whether or not certain plays that result in discarding/burning resources are worth it. Through training, the individual cards should learn at what point they are best used and incorporate this fact into its "report" that is sent to the director network. LSTM should help the player consider all stages of the game in its analysis of whether or not playing a card at a certain time is worth it.



Conclusion

The problems that mastering *Pokémon TCG* presents for an AI are not unique to the game. Whether it be analyzing the nearly infinite sample space of possible news articles and how they relate to one another in an attempt to separate real and fake news, determining the sequence of steps in a complicated problem in order to maximize the chances of success, or tackling many related, but vastly different, tasks with the same network architecture and no new training required, research into this approach to playing the complicated game of *Pokémon* will advance machine learning and AI research one step closer towards creating Artificial General Intelligence.

About the Author

Charlie Lockyer is a 17-year-old high school senior from Owings Mills, Maryland, who is deeply interested in computer science, artificial intelligence, and strategy games. Charlie also competes as a world-class Pokémon TCG player, having finished consistently well on the international circuit and in 18th place at the 2019 Pokémon TCG World Championships.

Goals for the Future

Charlie has built a global group of software developers with an interest in this problem, many of which work for companies like Google and Amazon, who have acted as mentors throughout the conceptualization phase of this project. However, both Charlie and his group have realized that their limited time and resources will halt further developments of this project. They hope to see this project come to fruition sooner rather than later and are looking for collaborators to assist in progressing towards an AI that can play the complicated game of Pokémon TCG.

